

ChimeraLM: a language model enables accurate structural variant detection in whole-genome amplified long-read sequencing

Yangyang Li^{1†}, Qingxiang Guo^{1†}, Rendong Yang^{1,2*}

¹Department of Urology, Northwestern University Feinberg School of Medicine, 303 E Superior St, Chicago, 60611, IL, USA.

²Robert H. Lurie Comprehensive Cancer Center, Northwestern University Feinberg School of Medicine, 675 N St Clair St, Chicago, 60611, IL, USA.

*Corresponding author(s). E-mail(s): rendong.yang@northwestern.edu;

Contributing authors: yangyang.li@northwestern.edu;

qingxiang.guo@northwestern.edu;

[†]These authors contributed equally to this work.

Abstract

Single-cell genomic analysis relies on whole genome amplification (WGA) to generate sufficient DNA for sequencing, yet this process introduces chimera artifacts that manifest as false-positive structural variation (SV) and compromise downstream analyses. Here we present ChimeraLM, an interpretable genomic language model (GLM) that identifies WGA-induced chimera artifacts directly from sequence information. ChimeraLM is trained on matched WGA and bulk long-read sequencing from the same sample, using bulk support to label chimeric reads as amplification-induced artifacts or genuine genomic events. To capture long-range dependencies in variable-length reads, the model integrates Hyena operators with attention pooling. Evaluated on matched WGA and bulk nanopore datasets, ChimeraLM eliminated $\sim 90\%$ of chimeric reads from WGA data, restoring chimeric read proportion to near-bulk levels, whereas existing approaches achieved at most an 8% reduction. Benchmarked against a multi-platform gold-standard SV set, ChimeraLM removed 91–98% of unsupported SV calls from WGA datasets while retaining 75–87% of gold-standard SVs; against platform-matched bulk long-read call sets, it removed 91–99% of unsupported calls while retaining 75–86% of bulk-supported SVs. ChimeraLM further

restored SV type distributions to near-bulk profiles, reducing the inversion fraction of WGA call sets from 94% and 57% to 12% and 5% on the two platforms, thereby largely removing the characteristic inversion bias observed in unprocessed WGA data. Attention-based interpretation indicates that ChimeraLM concentrates classification evidence at chimeric junctions, demonstrating its ability to learn biologically interpretable features. ChimeraLM provides a general approach for suppressing amplification-induced artifacts, enabling more reliable single-cell SV analysis across long-read platforms.

Keywords: Whole Genome Amplification, Single Cell, Genomic Language Model, Structural Variation

Main

Single-cell and low-input genomics have transformed our ability to resolve biological heterogeneity, enabling the discovery of rare cell states and the reconstruction of clonal evolution in cancer and development [1–3]. However, the limited DNA input (on the order of picograms per cell) makes comprehensive genome-wide profiling technically challenging [4, 5]. Whole genome amplification (WGA) therefore remains a prerequisite for high-coverage sequencing [6–8], yet it introduces systematic errors that compromise genomic fidelity, particularly for structural variation (SV) detection [9–11].

A major source of error in WGA is amplification-induced chimera formation. During this process, highly processive polymerases such as phi29, which is used in multiple displacement amplification (MDA), can switch templates and join discontinuous genomic loci into a single molecule [9–13]. As a result, WGA-based sequencing often produces chimeric reads that constitute a substantial fraction of the data [9]. These artificial sequences frequently create alignment patterns that closely resemble those generated by genuine SVs, including translocations and inversions [10]. Consequently, SV callers that rely on alignment-based signals (e.g., split-read and supplementary alignments) and coverage-derived evidence often misinterpret these amplification artifacts as true rearrangements, inflating false positives and distorting SV spectra [14–22]. This problem is particularly consequential for WGA-based long-read sequencing, which is otherwise well suited for resolving complex SVs at single-cell resolution [23, 24].

Distinguishing genuine genomic rearrangements from WGA-induced chimera artifacts remains a major computational challenge. Existing computational approaches detect amplification chimeras with handcrafted rules applied to alignments. SACRA [25], developed for Pacific Biosciences (PacBio) reads of MDA-amplified phage metagenomes, aligns reads against one another, locates positions at which many reads stop aligning while others align through, and splits reads at these candidate junctions. 3rd-ChimeraMiner [13], also developed on PacBio MDA data, identifies reads with multiple reference alignments and classifies them into chimera categories defined by the orientation and genomic distance of adjacent segments. CADECT [11], designed for Oxford Nanopore Technologies (ONT) MDA data, cuts each read into

sliding windows, aligns the windows against each other and flags reads whose windows align to one another as concatemers. Each of these methods therefore targets one structural signature of chimera formation (a junction detectable by read-versus-read alignment, a discordant reference alignment pattern, or an internal repeat), with thresholds tuned to a single platform, and none of them uses the nucleotide content around the junction. Because WGA chimeras are heterogeneous, comprising intra- and inter-molecular joins, inverted and direct configurations, and junctions with or without microhomology [9, 10], signature-specific heuristics are sensitive to platform- and protocol-specific variation and leave the chimera classes they were not designed for untouched; moreover, by construction they cannot capture sequence-level patterns or long-range dependencies within reads. As a result, low-input long-read sequencing remains difficult to deploy in settings where high precision is essential, including somatic mosaicism profiling [26] and validation of CRISPR off-target effects [27].

To address this challenge, we present ChimeraLM, an interpretable genomic language model (GLM) for identifying and filtering WGA-induced artifacts at the single-read level. Unlike existing approaches that rely on handcrafted rules derived from read alignments or sequence-level features [11, 13, 25], ChimeraLM formulates artifact detection as a sequence-modeling task and learns discriminative features directly from raw reads [28]. Building on advances in DNA foundation models [29–32], it captures latent motifs and structural dependencies that generalize across long-read sequencing platforms provided by ONT. On ONT WGA datasets, ChimeraLM reduces chimeric reads by ~90% while preserving 75–87% of gold-standard SVs and 75–86% of platform-matched bulk-supported SVs, improving SV validation rates by 3.2- to 26.6-fold and mitigating WGA-induced biases in SV type distributions. Together, ChimeraLM provides an effective and interpretable filter for WGA long-read data, enabling robust SV discovery in single-cell and low-input genomics.

Results

Overview of ChimeraLM workflow and model architecture

ChimeraLM operates as a post-alignment filtering module within the single-cell long-read sequencing pipeline (Fig. 1a). After base calling and alignment to the reference genome, the resulting read set typically contains a mixture of true chimeric reads that arise from authentic genomic rearrangements and artificial chimeras introduced during WGA. ChimeraLM evaluates all reads with chimeric alignments before variant calling. For each read, the model determines whether the observed chimera reflects a genuine genomic event or a WGA-induced artifact. This binary classification enables selective removal of artificial chimeric reads while preserving true rearrangements, which improves the accuracy and sensitivity of downstream SV analyses.

To construct a robust supervised training set, we generated WGA long-read sequencing data from the human prostate cancer cell line PC3 using the ONT PromethION platform. For ground-truth calibration, we acquired three matched bulk long-read datasets from unamplified genomic DNA across diverse technologies: ONT PromethION, ONT MinION, and PacBio Sequel II. Leveraging these references, we established a bulk-supported labeling framework by cross-referencing each WGA

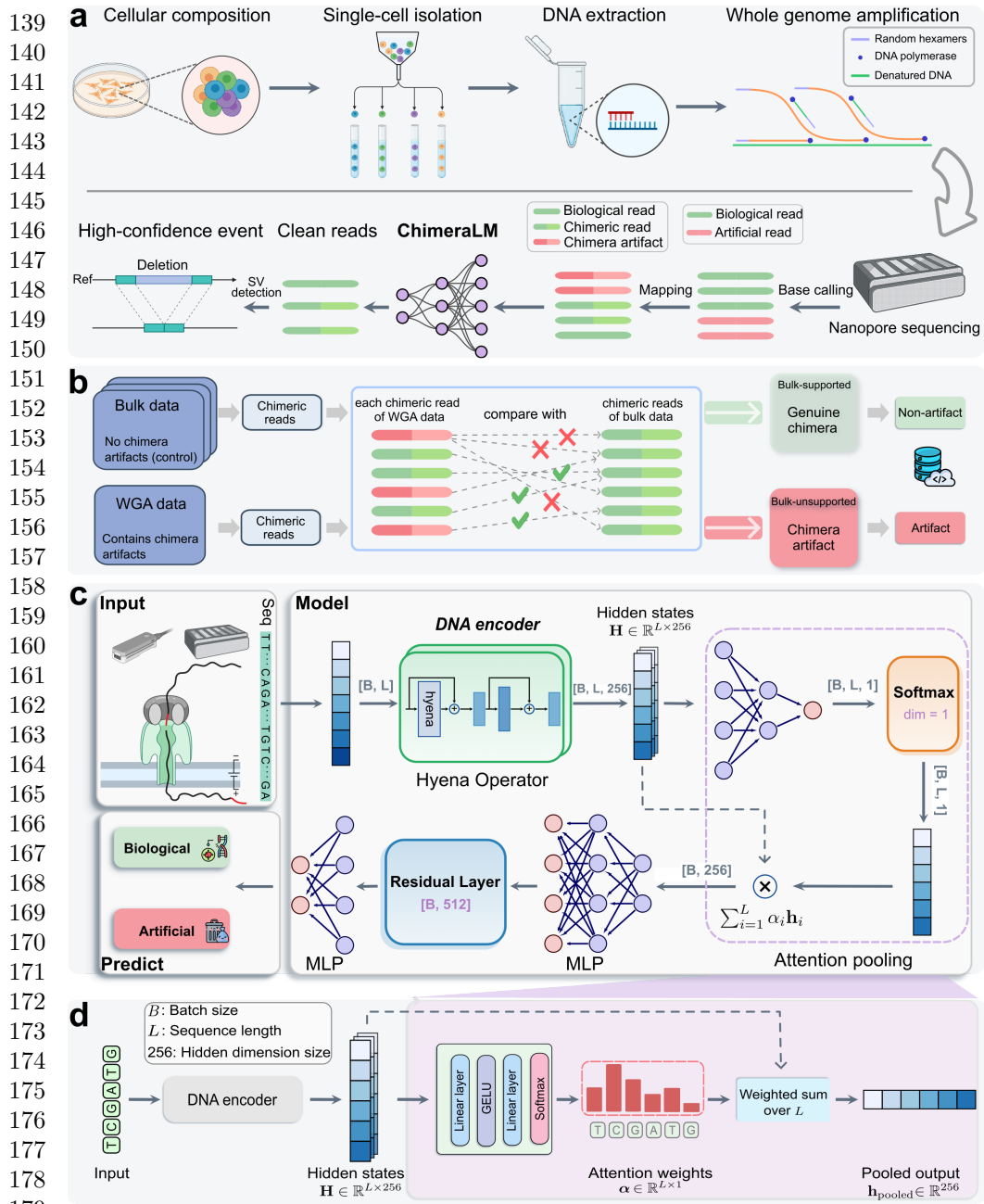


Fig. 1 ChimeraLM workflow and architecture for detecting WGA artifacts. (a) Single-cell genomic workflow and ChimeraLM integration. Single cells are isolated, followed by DNA extraction and WGA. During amplification, WGA-induced chimeric artifacts (red) are generated alongside genuine chimeric reads (green). After base calling and mapping, ChimeraLM classifies reads with chimeric alignments as genuine events or WGA-induced artifacts, enabling downstream SV detection on filtered data. (b) Bulk-supported label generation. Chimeric reads from WGA data are compared against bulk sequencing from the same cell line. Reads with bulk-supported alignment structures are labeled as genuine events (green); reads with no bulk match are labeled as WGA-induced artifacts (red). (c) ChimeraLM architecture. Input DNA sequences (batch size B , sequence length L) are tokenized at single-nucleotide resolution and encoded into hidden states $H \in \mathbb{R}^{L \times 256}$ through DNA encoder (HyenaDNA [29]). Hyena operators capture long-range dependencies. Attention pooling aggregates position-specific features, and multilayer perceptron (MLP) layers with residual connections process pooled representations for binary classification of genuine events and WGA-induced artifacts. (d) Attention pooling mechanism. Attention weights $\alpha \in \mathbb{R}^{L \times 1}$ are computed through linear layers with GELU activation and softmax normalization, assigning importance scores to each position. The weighted sum produces a fixed-dimensional representation $h_{\text{pooled}} \in \mathbb{R}^{256}$. Created with BioRender.com.

chimeric read against the chimeric alignment structures identified in the unamplified bulk data (Methods; Fig. 1b; Extended Data Fig. 1a). Under this classification scheme, WGA reads with alignment architectures corroborated by bulk evidence were labeled as genuine genomic events, while those lacking bulk support were classified as WGA-induced artifacts. To evaluate the model’s ability to generalize across sequencing hardware, we generated an independent WGA dataset on the MinION platform, which was reserved exclusively for testing (Extended Data Fig. 1a).

This labeling procedure classified 12,963,576 chimeric reads from the WGA PromethION dataset into two groups (genuine events and WGA-induced artifacts) based on bulk support (Extended Data Fig. 1b). Among these, 12,670,396 reads (97.7%) showed no matching alignment structures in any bulk dataset and were labeled as WGA-induced artifacts. The remaining 293,180 reads (2.3%) had matching structures in at least one bulk dataset, indicating they represent genuine genomic events rather than amplification artifacts, and were labeled as genuine chimeric reads. To construct a balanced training dataset, we retained all 293,180 genuine chimeric reads and randomly subsampled an equal number of WGA-induced artifacts. We further added 178,748 chimeric reads sampled from the bulk datasets to the genuine-event set, expanding the diversity of bulk-supported chimeric alignment structures used for training. The final labeled dataset comprised 765,108 reads and was split into training (70%), validation (20%), and test (10%) sets using stratified sampling (Extended Data Fig. 1a).

To model these labeled reads, ChimeraLM needs to process long, variable-length DNA sequences at single-nucleotide resolution (Fig. 1c). We therefore built ChimeraLM on HyenaDNA [29], a genomic foundation model pre-trained on diverse DNA sequences. Each read is tokenized at nucleotide resolution and encoded by Hyena operators [33], which capture long-range sequence context without splitting the input. The encoder produces a sequence of hidden states across the full read. To obtain a fixed-length representation for classification, ChimeraLM uses an attention-pooling module that learns position-specific weights and computes a weighted sum over the hidden states (Fig. 1d). The pooled representation is then passed through residual MLP blocks, and a final softmax outputs the probability that a read reflects a genuine event versus a WGA-induced artifact.

ChimeraLM achieves high accuracy and reduces artifacts to near-bulk levels across platforms

We first benchmarked ChimeraLM using the held-out test split (10%) derived from the bulk-supported labeled dataset (Extended Data Fig. 1a). This test set consists of chimeric reads labeled as genuine events or WGA-induced artifacts based on whether their chimeric alignment structures are supported by matched bulk sequencing. ChimeraLM achieved an F1 score of 0.81, with 0.95 recall and 0.70 precision (Fig. 2a). The high recall indicates that most WGA-induced artifacts are correctly identified for removal, which is important for limiting downstream false-positive SV calls, while the precision confirms that most reads flagged as artifacts are true amplification-induced chimeras rather than genuine genomic events.

Because ChimeraLM processes at most 32,768 bp per read, we examined how often this limit is reached and whether accuracy depends on read length (Extended Data

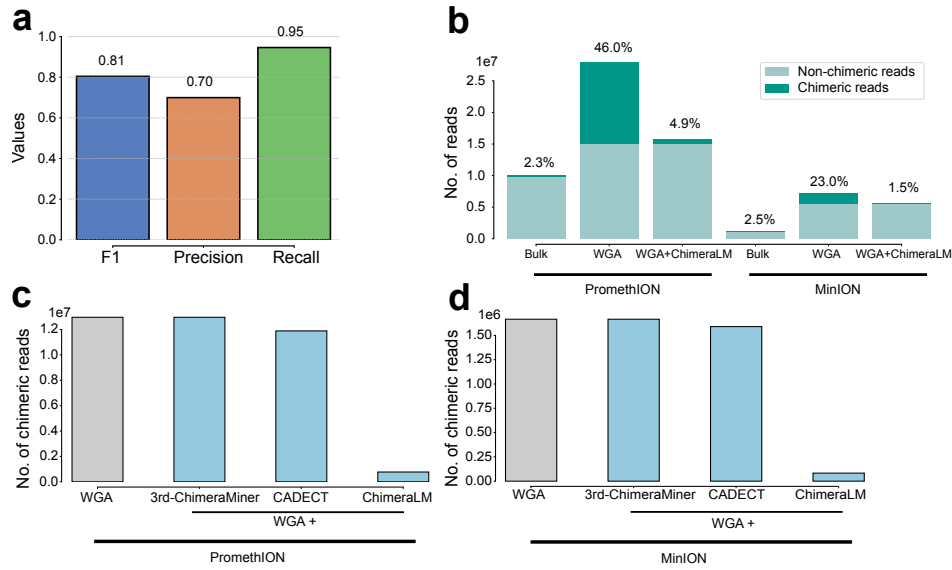


Fig. 2 ChimeraLM accurately identifies and removes WGA-induced chimeric artifacts. (a) Classification performance on held-out test data. ChimeraLM achieves recall of 0.95, precision of 0.70, and F1 score of 0.81. (b) Chimeric read reduction across sequencing platforms. Stacked bars show proportions of chimeric (dark teal) and non-chimeric (light teal) reads in bulk, WGA, and ChimeraLM-filtered samples. ChimeraLM reduces chimeric read frequencies from 46.0% to 4.9% (PromethION) and from 23.0% to 1.5% (MinION), approaching bulk levels (2.3% and 2.5%, respectively). (c,d) Benchmarking against existing methods on PromethION (c) and MinION (d). The gray bar indicates the total number of chimeric read on unfiltered WGA data. The blue bar represents the total number of chimeric reads remaining after filtering by each method. ChimeraLM achieves approximately 90% reduction in chimeric reads on both platforms, while 3rd-ChimeraMiner shows no detectable reduction and CADECT shows 8.3% and 4.6% reduction on PromethION and MinION, respectively. SACRA failed to complete due to memory exhaustion (> 500 GB RAM required).

Fig. 2). Chimeric reads in WGA data are short (median 2,100 bp on PromethION, 1,469 bp on MinION); only 0.009% exceeded 32,768 bp, and all but two of these retained at least one junction within the window (Extended Data Fig. 2a). Accuracy nevertheless varied with length: on WGA-derived test reads, precision rose from 0.69 (≤ 2 kb) to ≥ 0.99 (> 8 kb) while recall fell from 0.98 to 0.31 (16–32 kb), so rare long reads (> 16 kb, 0.26% of chimeric reads) are classified conservatively (Extended Data Fig. 2c). This likely reflects the bulk-sampled genuine reads in the training set, which are far longer (median 14.4 kb) than WGA reads, so that length carries label information; consistently, removed chimeric reads were shorter than retained ones (median 2,067 versus 3,087 bp on PromethION; Extended Data Fig. 2d), whereas non-chimeric reads, the majority of a WGA library, are not evaluated and remain unchanged.

We next examined whether applying ChimeraLM filtering would restore chimeric read rates in full PC3 WGA datasets toward the levels observed in bulk sequencing across both PromethION and MinION platforms (Fig. 2b). Bulk sequencing established low baseline chimeric read rates of 2.3% (PromethION) and 2.5% (MinION).

In contrast, WGA increased the chimeric read fraction to 46.0% and 23.0%, respectively. After ChimeraLM filtering, chimeric content dropped to 4.9% on PromethION and 1.5% on MinION, corresponding to 10- to 15-fold reductions, while retaining 15.8 million and 5.6 million reads. These post-filtering rates approach the corresponding bulk baselines, indicating effective removal of WGA-induced artifacts while preserving genuine signal.

We compared ChimeraLM against SACRA [25], 3rd-ChimeraMiner [13], and CADECT [11], existing tools for detecting amplification-induced chimeras (Fig. 2c,d). ChimeraLM reduced chimeric reads by ~90% on both platforms, outperforming CADECT (8.3% and 4.6% reduction on PromethION and MinION, respectively), while 3rd-ChimeraMiner showed no detectable reduction. SACRA could not be evaluated due to out-of-memory errors even with 500 GB RAM.

The MinION results further provide an independent test of model generalization, as this MinION WGA dataset was not used during training. ChimeraLM was trained exclusively on PromethION WGA data, yet achieved comparable chimeric read reduction on MinION. This cross-platform generalization indicates that ChimeraLM captures sequence-level features intrinsic to WGA-induced artifacts rather than platform-specific signatures, supporting its potential applicability to additional long-read and potentially short-read sequencing technologies.

To test generalization beyond the training cell line and laboratory, we applied the unchanged model to public single-cell WGA data of the lung adenocarcinoma line HCC78 generated by another group [34]: one MDA cell (R10.4 flow cell), one multiple annealing and looping-based amplification cycles (MALBAC) cell (R9.4.1) and two unamplified bulk libraries used as reference (Methods; Extended Data Fig. 3). ChimeraLM removed 81.7% of chimeric reads from the MDA library, lowering its chimeric fraction from 66.6% to 26.8% (HCC78 bulk: 21.0% and 7.9%), and 87.3% from the MALBAC library (6.8% to 0.9%), with recall of 0.82 and 0.87 for bulk-unsupported reads. Artifact removal thus transfers across cell line, laboratory, flow-cell chemistry and amplification method without retraining.

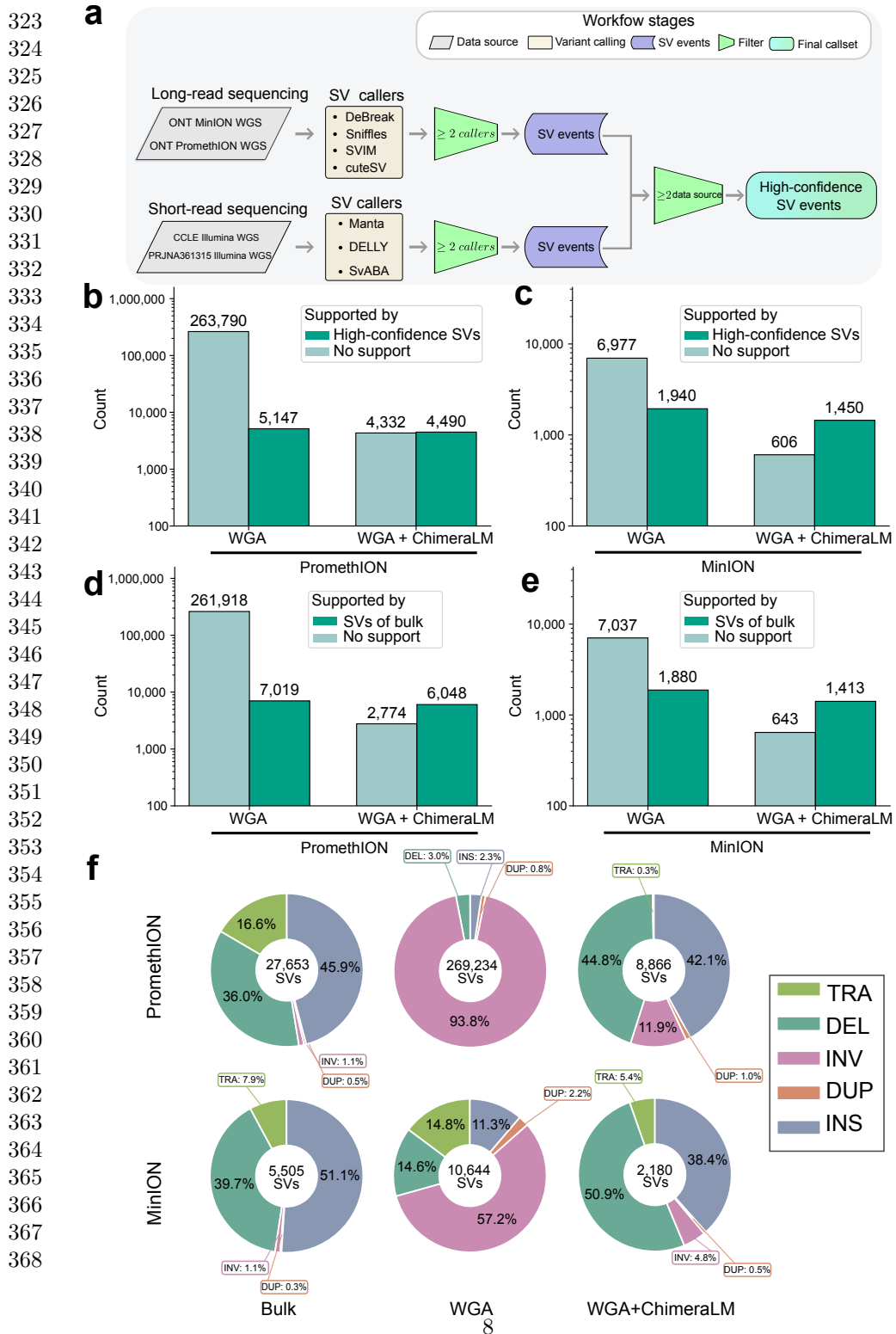


Fig. 3 ChimeraLM improves structural variant detection accuracy. (a) Construction of a high-confidence SV reference dataset from bulk PC3 sequencing. Four bulk datasets were integrated: ONT MinION Mk1C, ONT PromethION P2, the CCLE Illumina whole genome sequencing (WGS) dataset, and the PRJNA361315 Illumina WGS dataset. SVs were called independently within each dataset using multiple callers, and events supported by ≥ 2 callers per dataset were retained. SVs were then compared across datasets, and events observed in ≥ 2 of the four bulk datasets were designated as gold-standard SVs. (b,c) SV validation against the gold-standard reference for PromethION (b) and MinION (c). Grouped bars show, on a shared logarithmic scale, the number of SV calls supported by the gold standard (dark teal) or unsupported (light teal); the unsupported-to-supported ratio is given below each group. Only calls with at least three supporting reads (Sniffles2 default) are included. (d,e) SV validation against platform-matched long-read bulk sequencing for PromethION (d) and MinION (e), capturing true long-read SVs that may not be represented in the multi-platform reference. Grouped bars show SV calls supported by the platform-matched long-read bulk data (dark teal) or unsupported (light teal), as in (b,c). (f) Distribution of SV types for PromethION and MinION datasets across bulk samples, unfiltered WGA, and WGA with ChimeraLM filtering, for calls with at least three supporting reads. Unfiltered WGA exhibits an excess of inversion (INV) events, which is substantially reduced following ChimeraLM filtering. Other SV classes, including translocation (TRA), deletion (DEL), duplication (DUP), and insertion (INS), are also shown.

ChimeraLM reduces unsupported structural variant calls

Accurate SV detection from single cells is essential for characterizing genomic diversity and disease mechanisms. However, WGA-induced chimeric artifacts can be misinterpreted as genuine SVs, potentially leading to incorrect biological inferences. To assess the impact of ChimeraLM on SV calling, we compared SV callsets generated from unfiltered WGA reads with those obtained after applying ChimeraLM filtering (WGA + ChimeraLM). Both callsets were evaluated against two complementary bulk-derived references: (i) a stringent gold-standard SV set derived from bulk PC3 DNA through cross-dataset consensus, and (ii) platform-matched long-read bulk SV callsets used as platform-specific references.

We first constructed a high-confidence gold-standard SV set from bulk PC3 DNA by integrating four independent sequencing datasets: ONT PromethION, ONT MinION, and two Illumina short-read WGS datasets from the Cancer Cell Line Encyclopedia (CCLE Illumina WGS [35]) and from a previously published study (PRJNA361315 Illumina WGS [36]) (Fig. 3a; Extended Data Table 1). SVs were called separately within each dataset using multiple SV callers. Events supported by at least two callers within a dataset were retained, and only SVs observed in at least two of the four datasets were designated as gold-standard events. Of the 17,153 events in this two-of-four set, 73.8% are shared only by the two long-read datasets (Extended Data Fig. 4); because only two datasets are long-read, a three-of-four rule would require short-read confirmation, keep just 3,029 events (17.7%) and exclude variant classes that short reads detect poorly [14], so the two-of-four set is complemented by the platform-matched long-read references below (Fig. 3d,e).

Call sets were restricted to SVs with at least three supporting reads, the Sniffles2 default (Methods). Relative to this stringent gold standard, unfiltered WGA produced a large excess of unsupported SVs. On PromethION, WGA yielded 268,937 SV calls, of which only 5,147 (1.9%) overlapped gold-standard events, an unsupported-to-supported ratio of 51:1. After ChimeraLM filtering, total calls decreased to 8,822 (a 96.7% reduction); unsupported calls fell from 263,790 to 4,332 (98.4% removed) while 4,490 gold-standard events were retained (87.2% retention), so that supported calls outnumbered unsupported calls (ratio 0.96:1) and the validation rate rose from 1.9% to 50.9% (26.6-fold) (Fig. 3b). On MinION, total calls decreased from 8,917 to 2,056 (76.9% reduction); unsupported calls fell from 6,977 to 606 (91.3% removed) while gold-standard-supported events decreased from 1,940 to 1,450 (74.7% retention), raising the validation rate from 21.8% to 70.5% (3.2-fold; ratio 3.6:1 to 0.42:1) (Fig. 3c). The smaller fold change on MinION reflects the lower artifact burden of its call set at this read-support threshold (21.8% of calls already validated before filtering, versus 1.9% on PromethION).

Because the gold standard is intentionally stringent and may exclude true SVs detectable only in long-read data, we next performed platform-matched validation using long-read bulk sequencing from the same platform (Fig. 3d,e). This analysis provides a platform-specific estimate of recall and reduces bias introduced by the strict gold-standard definition. ChimeraLM increased validation rates from 2.6% to 68.6% on PromethION (26.3-fold) and from 21.1% to 68.7% on MinION (3.3-fold), removing 98.9% and 90.9% of bulk-unsupported calls while retaining 86.2% (6,048 of 7,019) and

75.2% (1,413 of 1,880) of bulk-supported events, respectively (Fig. 3d,e). Together, these results show that ChimeraLM removes the vast majority of unsupported SV calls while preserving most bulk-supported variants across platforms, leaving call sets in which supported SVs outnumber unsupported ones.

We next characterised the unsupported calls that remain after ChimeraLM filtering relative to the gold standard and compared them with the supported calls in the same call sets (Extended Data Fig. 5). Residual unsupported calls were not dominated by a single class: on PromethION they comprised 39.5% deletions, 34.9% insertions, 24.0% inversions and 1.7% duplications, were small (median 189 bp; 70.2% <500 bp) and weakly supported (median 4 reads; 65.3% with ≤ 5 reads), whereas supported calls had a median of 10 supporting reads. Inversions were the one diagnostic feature: 1,039 residual unsupported calls were inversions, compared with a single inversion among the 4,490 supported calls (MinION: 100 versus 0). In contrast, the residual deletions and insertions resembled supported calls in size, consistent with a mixture of low-support artifacts and genuine variants that are absent from the stringent gold standard (see Extended Data Fig. 6). Raising the minimum read support further reduces residual calls at a cost in sensitivity: on PromethION, requiring at least five supporting reads removed 56.3% of residual unsupported calls but also 20.1% of supported calls (unsupported-to-supported ratio 0.96:1 to 0.53:1), and requiring ten reads removed 80.3% versus 48.6%; on the shallower MinION data stricter thresholds removed both classes at similar rates.

ChimeraLM mitigates WGA-induced SV type biases

Amplification artifacts can profoundly distort the apparent spectrum of SVs, complicating biological interpretation. To quantify these effects, we compared SV-type compositions across bulk, unfiltered WGA, and ChimeraLM-filtered datasets generated on both ONT PromethION and MinION platforms (Fig. 3f). Bulk datasets displayed SV profiles dominated by deletions and insertions (81.9% and 90.8% combined on PromethION and MinION, with inversions at 1.1% on both), whereas WGA datasets exhibited a striking inflation of inversions, reaching 93.8% on PromethION and 57.2% on MinION among calls with at least three supporting reads. This inversion-heavy signature mirrors previous observations from single-cell long-read sequencing using PacBio and droplet-based MDA [22], suggesting that many of these events arise from amplification-induced chimeras rather than genuine genomic rearrangements.

To directly assess which SV types are preferentially associated with amplification artifacts, we examined SV calls supported exclusively by reads classified as WGA-derived chimeras. These artifact-supported events were overwhelmingly enriched for inversions, comprising 88.4% on PromethION and 92.4% on MinION (Extended Data Fig. 7). Smaller fractions of deletions, duplications, and insertions were also present, indicating that WGA artifacts inflate not only inversion calls but generate false positives across multiple SV categories.

Applying ChimeraLM largely removed these biases. The inversion fraction decreased to 11.9% on PromethION and 4.8% on MinION, while deletions and insertions rose to 86.9% and 89.3% of calls, close to bulk (Fig. 3f); on MinION the excess

of translocation-type (breakend) calls was likewise reduced from 14.8% to 5.4% (bulk 461
7.9%). The residual inversion fraction remains above the bulk level, and the remaining 462
inversions are almost entirely unsupported by bulk sequencing (Extended Data Fig. 5). 463
By selectively removing artifact-supported events, ChimeraLM suppresses the domi- 464
nant inversion inflation and reduces spurious calls across SV types, shifting SV-type 465
distributions toward bulk profiles and improving the accuracy and interpretability of 466
single-cell SV analyses [22]. 467

468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506

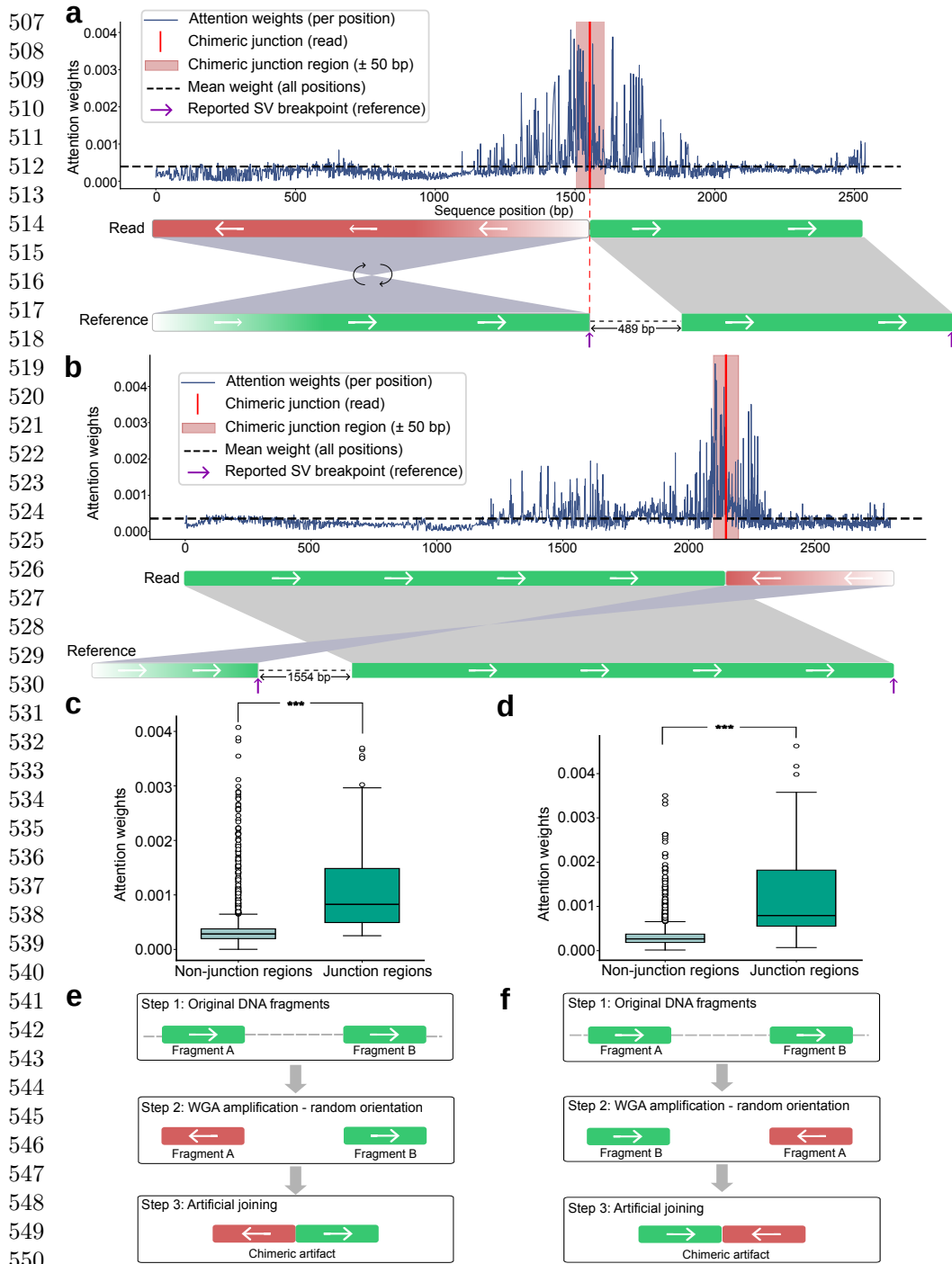


Fig. 4 ChimeraLM attention weights are enriched at chimeric junction regions. (a,b) Attention weight profiles for two representative chimeric reads exhibiting distinct junction configurations. Upper panels show per-position attention weights (blue) with the mean attention across the read indicated by a dashed line. Red vertical lines mark inferred chimeric junction positions, and pink shading denotes the junction-centered region (± 50 bp). Lower panels display read-level alignments, highlighting orientation transitions at the junctions (green, forward orientation; red, reverse-complemented orientation). (c,d) Quantitative comparison of attention weights between junction and non-junction regions. Junction-centered windows show significantly elevated attention weights relative to non-junction regions ($P = 5.3 \times 10^{-14}$ and $P = 6.8 \times 10^{-15}$; Wilcoxon rank-sum test). (e,f) Schematic illustration of WGA-induced chimera formation. During amplification, DNA fragments originating from distant genomic loci can be amplified in either orientation, joining them into a single molecule with discordant orientations, producing inversion-like alignment signatures. The two examples illustrate forward-to-reverse and reverse-to-forward orientation transitions.

Attention visualization reveals interpretable classification features

We examined whether ChimeraLM’s attention weights provide an interpretable signal by focusing on mechanistically relevant regions of chimeric reads, particularly the junctions during WGA (Fig. 4). We inspected two representative chimeric reads exhibiting distinct junction configurations (Fig. 4a,b). In both cases, attention remained relatively flat across most positions but formed sharp, concentrated peaks at the inferred junction regions. These peaks aligned with the read-level breakpoint separating two genomic loci and coincided with an orientation transition between adjacent alignment segments.

To quantify this effect, we compared attention weights within junction-centered windows (± 50 bp) against weights from non-junction regions (Fig. 4c,d). Junction windows showed significantly higher attention weights (Wilcoxon rank-sum test, $P = 5.3 \times 10^{-14}$ and $P = 6.8 \times 10^{-15}$), indicating that ChimeraLM preferentially emphasizes sequence context proximal to chimeric junctions.

This attention enrichment is consistent with the expected structure of WGA-induced chimeras (Fig. 4e,f): DNA fragments from distant loci can be amplified in either orientation and subsequently joined, generating junctions with discordant orientations. Together, these results suggest that ChimeraLM’s attention peaks provide a mechanistically interpretable signal that concentrates classification evidence to junction-proximal sequence positions within individual reads.

Discussion

WGA enables genomic analysis from single cells and other low-input samples, but it also introduces chimeric artifacts that compromise SV detection. ChimeraLM addresses this challenge by classifying chimeric reads as genuine events or WGA-induced artifacts directly from read information, and removing artifacts before variant calling, rather than attempting to correct artifact-driven calls post hoc. Across ONT platforms, ChimeraLM improved data quality at the read and variant levels: it reduced chimeric reads by $\sim 90\%$ while retaining 75–87% of gold-standard SVs and 75–86% of platform-matched bulk-supported SVs, and it increased the fraction of supported SV calls by 3.2- to 26.6-fold. Notably, performance generalized from PromethION (used for training) to MinION without platform-specific retraining, suggesting that ChimeraLM captures properties shared by WGA-induced artifacts rather than instrument-specific signatures.

In comparison, existing methods showed limited effectiveness on our ONT WGA datasets. Two tools originally developed and primarily evaluated on PacBio data (SACRA and 3rd-ChimeraMiner) [13, 25] either failed to complete under our evaluation setting (SACRA, > 500 GB RAM) or showed no detectable reduction (3rd-ChimeraMiner), highlighting poor cross-platform generalization. CADECT [11], which was designed for ONT data, achieved only 8.3% and 4.6% reduction on PromethION and MinION, respectively. CADECT detects concatemers via sliding-window self-alignment, an effective strategy for repeat-like structures with internal sequence similarity, but it is not designed to capture the broader diversity of WGA-induced

599 chimeras. Together, these comparisons suggest that rule-based or subtype-specific
600 heuristics do not comprehensively address amplification-induced chimeras and moti-
601 vate learning-based models that can extract discriminative sequence signatures
602 without imposing predefined structural assumptions.

603 ChimeraLM also illustrates the value of deep learning for quality-control problems
604 where conventional alignment- and coverage-derived criteria provide limited resolu-
605 tion. [11, 13, 25, 28]. By learning directly from sequence, ChimeraLM discovers subtle
606 compositional and structural features that separate genuine events from amplification
607 artifacts. The model also offers interpretability through attention visualization: atten-
608 tion weights concentrate at junction regions where template switching joins discordant
609 loci, validating the biological relevance of the learned features. These methodological
610 advances have direct implications for single-cell genomics, where high false-positive
611 rates in WGA data have constrained robust characterization of chromosomal instabil-
612 ity, clonal evolution, and SV burden [20, 22, 37]. By improving the signal-to-noise ratio
613 and clarifying SV-type spectra that are otherwise distorted by amplification artifacts,
614 ChimeraLM enables more confident identification of genuine SVs, supporting studies
615 of cancer evolution, developmental biology, and somatic mosaicism where single-cell
616 resolution is essential [26, 27].

617 Several limitations warrant consideration. First, the current model processes reads
618 independently; integrating contextual features such as coverage or phasing informa-
619 tion may further enhance accuracy. Second, regarding computational resources, while
620 central processing unit (CPU) inference is feasible, graphics processing unit (GPU)
621 acceleration is recommended for processing large-scale datasets. Third, the training
622 labels inherit the incompleteness of bulk sequencing: a genuine chimeric read whose
623 structure is absent from all three bulk datasets is labelled as an artifact (Extended
624 Data Fig. 6), and the model may consequently remove reads of that kind; this error
625 reduces sensitivity for rare genuine events rather than creating false positives, and
626 its magnitude is bounded by the observed retention of bulk-supported SVs (75–87%).
627 We have shown that artifact removal transfers without retraining to an independent
628 cell line, laboratory, flow-cell chemistry (R9.4.1 and R10.4) and amplification method
629 (MALBAC); when a matched bulk dataset is available, fine-tuning the classifier on
630 bulk-supported reads of the sample under study is nevertheless a straightforward way
631 to adapt it to sample-specific rearrangements. Future work should extend validation to
632 further cell types, sequencing platforms (e.g., PacBio HiFi), and alternative WGA pro-
633 tocols, including linear amplification via transposon insertion (LIANTI) [5], primary
634 template-directed amplification (PTA) [19], and droplet-based MDA (dMDA) [38].
635 While the sequence-based approach suggests broad applicability, systematic validation
636 across amplification chemistries is needed to assess generalization limits and optimize
637 performance for specific protocols. Validation in primary tissue and tumour samples
638 remains an open task: evaluating a read-level artifact filter requires single-cell long-
639 read WGA data together with matched bulk long-read sequencing of the same sample,
640 and the few such datasets from human tissue published to date are limited. Because
641 tissue and tumour cells are more heterogeneous than a monoclonal cell line, whereas
642 the artifact profile is determined primarily by the amplification chemistry rather than
643 by the cell of origin, we expect ChimeraLM’s performance on such samples to track
644

its cross-cell-line and cross-chemistry behaviour, but this expectation should be tested as suitable data become accessible.

ChimeraLM acts on reads, not on calls, and is therefore complementary to, rather than a replacement for, caller-level filtering. Our characterisation of the calls that remain unsupported after filtering (Extended Data Fig. 5) suggests two practical steps for users who require higher specificity: inversion calls that lack orthogonal support should be treated with suspicion, because essentially no bulk-supported inversion survived in our WGA call sets while residual inversions accounted for 17–24% of unsupported calls; and a moderate read-support threshold (for example, at least five supporting reads) removes more than half of the residual unsupported calls on deeply sequenced samples at a cost of about one fifth of supported calls, whereas stricter thresholds increasingly discard supported calls as well, and on shallow data remove both classes at similar rates. Conversely, callers that aggregate evidence across cells or that incorporate coverage and phasing information may recover genuine events that single-cell read-level filtering alone cannot confirm.

More broadly, ChimeraLM illustrates the potential of GLMs for data quality control. As long-context architectures continue to advance [29], extending the model’s context window to handle megabase-scale inputs could enable artifact detection in more complex genomic structures. This framework could extend to other amplification-dependent technologies, such as cell-free DNA analysis, ancient DNA studies, and low-biomass metagenomics. Furthermore, attention-based interpretability opens opportunities for studying template-switching dynamics, potentially guiding the development of improved amplification protocols. In summary, ChimeraLM provides a practical and interpretable framework for enhancing long-read single-cell genomic fidelity, ensuring that downstream biological insights are derived from genuine SVs rather than technical artifacts.

Methods

Cell culture, single-clone preparation, and nanopore sequencing

Cell culture and single-clone establishment

PC3 prostate cancer cells (ATCC® CRL-1435™) were cultured in RPMI-1640 medium supplemented with 10% fetal bovine serum and 1% penicillin–streptomycin at 37 °C with 5% CO₂. To minimize biological heterogeneity, a monoclonal population was established by serial dilution in 96-well plates, ensuring that each culture originated from a single cell. Mycoplasma contamination was routinely tested and confirmed negative prior to DNA extraction.

DNA extraction and whole-genome amplification

From the monoclonal population, two types of DNA samples were prepared: a bulk (non-amplified) control and ten single-cell MDA-amplified genomes. Bulk high-molecular-weight DNA was extracted using the Monarch® HMW DNA Extraction Kit for Cells & Blood (New England Biolabs). Individual cells were isolated using

1CellDish-60 mm (iBioscience) and amplified using the REPLI-g Advanced DNA Single Cell Kit (Qiagen) following the manufacturer’s protocol. DNA concentration and fragment integrity were assessed with a Qubit 4 fluorometer and Agilent TapeStation (DNA 1000/5000 ScreenTape). Only samples meeting quality standards were used for library construction.

Nanopore library preparation and sequencing

Libraries were prepared using the ONT Ligation Sequencing Kit V14 (SQK-LSK114) and sequenced on MinION Mk1C or PromethION P2 Solo devices with R10.4.1 flow cells following the manufacturer’s genomic DNA workflow. Because all single-cell samples originated from the same monoclonal lineage, differences between amplified and bulk datasets primarily reflect MDA-induced artifacts rather than biological variation.

Basecalling and read processing

POD5 files were basecalled using Dorado v0.5.0 with the high-accuracy model dna_r10.4.1_e8.2.400bps_hac@v4.3.0 [39]. Reads with mean quality < 10 or length < 500 bp were removed. Adapters and concatemers were trimmed using Cutadapt v4.0 [40] in a two-pass, error-tolerant procedure. Filtered reads were aligned to the GRCh38.p13 reference genome using minimap2 v2.26 (map-ont preset) [41]. BAM files were sorted and indexed using SAMtools v1.16 [42]. Read-length and mapping statistics were computed using NanoPlot v1.46.1 [43]. All samples were processed using identical parameters.

Chimeric read identification

Chimeric reads were identified from BAM files using supplementary alignment (SA) tags. Reads were classified as chimeric if they (i) were mapped, (ii) contained an SA tag, (iii) were primary alignments (not secondary), and (iv) were not supplementary alignments themselves. This definition counts each chimeric read once using its primary alignment while excluding secondary/supplementary records, thereby avoiding double-counting and reducing ambiguity from low-confidence alignments. Reads lacking SA tags were classified as non-chimeric.

Training data construction

Data generation and sources

To construct the training dataset, we generated WGA and bulk sequencing data from PC3 cells. The WGA sample was amplified and sequenced on the PromethION P2 platform (ONT), while three independent bulk datasets were produced from non-amplified genomic DNA: bulk PromethION P2, bulk MinION Mk1c (ONT), and bulk PacBio. These bulk datasets represent authentic biological sequences free from amplification-induced artifacts. In contrast, WGA sequencing includes both genuine genomic reads and artificial chimeras introduced during the amplification process.

Ground truth annotation and class definition

Ground truth labels were established by systematically comparing chimeric reads from the WGA PromethION P2 dataset against those from the three bulk datasets. For each WGA chimeric read, all alignment segments—defined by their genomic start and end coordinates—were compared to the corresponding segments of bulk chimeric reads. A WGA read was labeled as biological if every segment matched at least one bulk chimeric read within a 1 kb positional tolerance, indicating that the structural configuration is also present in non-amplified DNA. Reads lacking any matching pattern across all bulk datasets were labeled as artificial chimeras, presumed to arise from the amplification process. Additional chimeric reads were randomly sampled from the bulk datasets and labeled as biological, as these reads originate from genuine genomic rearrangements such as true SVs. The final labeled dataset combined the annotated WGA PromethION P2 reads with the subsampled bulk chimeric reads and was subsequently partitioned into training, validation, and test sets as described below.

Cross-platform concordance of bulk SV calls

To estimate how incomplete each bulk reference is, SVs were called with Sniffles2 v2.5 on the bulk PromethION, MinION and PacBio datasets, retained if supported by at least three reads, and merged across platforms with OctopusSV v0.2.3 (`--union --mode sample`). For the depth titration, the bulk PromethION BAM (20.8×) was down-sampled with `samtools view -s` to 15×, 10×, 5× and 2.37× (the depth of the bulk MinION dataset); Sniffles2 was re-run on each down-sampled BAM with identical parameters and the resulting calls were merged with the full MinION and PacBio call sets. For each depth we report the number of PromethION calls and the fraction supported by at least one other platform.

Dataset partitioning and cross-platform validation

The combined labeled dataset, derived from WGA PromethION P2 and bulk sequencing data, was divided into training (70%), validation (20%), and test (10%) sets using stratified random sampling. These subsets were used respectively for model training, hyperparameter tuning, and performance evaluation on data from the same sequencing platform.

To evaluate cross-platform generalization, the complete WGA MinION Mk1c dataset was reserved. This dataset, generated on a different nanopore platform, was never used during model training or internal testing. This two-level evaluation design allowed us to test whether ChimeraLM captures general sequence features of amplification-induced chimeras rather than platform-specific artifacts.

Model architecture

DNA encoder

ChimeraLM employs the pre-trained HyenaDNA model [29] as its DNA encoder. This model was pre-trained on large-scale genomic data and provides robust sequence representations. DNA sequences are tokenized at single-nucleotide resolution, with each base (A, C, G, T, N) mapped to a unique integer token (7, 8, 9, 10, 11, respectively).

Special tokens include [CLS]=0, [PAD]=4, and others for sequence processing. Input sequences are truncated at 32,768 bp or padded to enable batch processing; reads longer than 32,768 bp are not discarded but are classified from their first 32,768 bp (Extended Data Fig. 2). The 32,768-bp limit is the context length of the pre-trained HyenaDNA-small backbone; it exceeds the length of all but 0.009% of chimeric reads in our data (Extended Data Fig. 2a), and HyenaDNA backbones pre-trained with 160-kb to 1-Mb contexts can be substituted without changes to the architecture should read lengths increase.

For a tokenized input sequence $\mathbf{x} \in \mathbb{Z}^L$, the HyenaDNA generates contextualized hidden representations:

$$\mathbf{H} = \text{HyenaDNA}(\mathbf{x}) \in \mathbb{R}^{L \times 256}$$

where $\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L)$ represents position-wise hidden states with dimension 256. The Hyena operators [33] efficiently capture both local sequence motifs and long-range dependencies essential for distinguishing biological sequences from chimeric artifacts.

Attention pooling

To aggregate variable-length sequence representations into fixed-size vectors, ChimeraLM implements attention-based pooling. For hidden states $\mathbf{H} \in \mathbb{R}^{L \times 256}$, attention weights are computed through a two-layer network:

$$\begin{aligned} \mathbf{e} &= \text{GELU}(\text{Linear}_{256 \rightarrow 256}(\mathbf{H})) \in \mathbb{R}^{L \times 256} \\ \mathbf{s} &= \text{Linear}_{256 \rightarrow 1}(\mathbf{e}) \in \mathbb{R}^{L \times 1} \\ \boldsymbol{\alpha} &= \text{softmax}(\mathbf{s}) \in \mathbb{R}^{L \times 1} \end{aligned}$$

The pooled representation is the weighted sum of hidden states:

$$\mathbf{h}_{\text{pooled}} = \sum_{i=1}^L \alpha_i \mathbf{h}_i \in \mathbb{R}^{256}$$

This mechanism assigns learned importance weights to each sequence position, enabling the model to focus on informative regions while accommodating natural variability in read lengths.

Classification head

The pooled representation is processed through a MLP with residual connections. The first layer expands dimensionality:

$$\mathbf{f}_1 = \text{Dropout}_{0.1}(\text{GELU}(\text{Linear}_{256 \rightarrow 512}(\mathbf{h}_{\text{pooled}}))) \in \mathbb{R}^{512}$$

Subsequent residual blocks with input $\mathbf{f}_{\text{in}} \in \mathbb{R}^{512}$ compute:

$$\mathbf{f}_{\text{out}} = \text{Dropout}_{0.1}(\text{Linear}_{512 \rightarrow 512}(\text{GELU}(\text{Linear}_{512 \rightarrow 512}(\mathbf{f}_{\text{in}})))) + \mathbf{f}_{\text{in}}$$

where the skip connection enables stable gradient flow during training. The final layer produces binary classification logits:

$$\mathbf{z} = [z_0, z_1] = \text{Linear}_{512 \rightarrow 2}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^2$$

where z_0 and z_1 represent logits for biological and artificial chimeric classes, respectively. During inference, the predicted class is $\hat{y} = \text{argmax}_{i \in \{0,1\}} z_i$.

Model summary

The complete ChimeraLM pipeline processes DNA sequences through: (1) single-nucleotide tokenization, (2) HyenaDNA backbone encoding to generate contextualized representations, (3) attention pooling to aggregate position-specific features, (4) MLP layers with residual connections to learn classification features, and (5) binary classification output. The entire model with $\sim 4.2\text{M}$ trainable parameters is trained end-to-end using labeled data.

Model training and optimization

Training configuration

ChimeraLM was trained using PyTorch [44] and PyTorch Lightning [45] frameworks. Input sequences were tokenized using the tokenizer with maximum sequence length of 32,768 bp. Sequences longer than this threshold were truncated; shorter sequences were padded to enable batch processing. Training employed mixed-precision computation (bf16) to accelerate training while maintaining numerical stability.

Optimization procedure

We used the AdamW optimizer [46] with learning rate $\eta = 1 \times 10^{-4}$ and weight decay $\lambda = 0.01$. AdamW implements adaptive learning rates with decoupled weight decay, combining the benefits of Adam optimization with proper L2 regularization. A ReduceLROnPlateau scheduler dynamically adjusted the learning rate based on validation loss, reducing it by a factor of 0.1 when no improvement occurred for 10 consecutive epochs. Early stopping with patience of 10 epochs prevented overfitting by terminating training when validation performance plateaued. A fixed random seed (12345) ensured reproducibility across training runs.

The training objective used cross-entropy loss for binary classification. For a training example with class label $y \in \{0, 1\}$ and model logits $\mathbf{z} = [z_0, z_1]$, the loss is:

$$\mathcal{L}(\mathbf{z}, y) = -\log \left(\frac{\exp(z_y)}{\exp(z_0) + \exp(z_1)} \right) = -z_y + \log(\exp(z_0) + \exp(z_1))$$

where z_0 and z_1 represent logits for biological and artificial chimeric classes, respectively.

Training implementation

Training used batch size of 16 sequences with 30 parallel data loading workers. GPU acceleration was employed for efficient processing, with training typically requiring

55 hours. Model checkpointing saved the best-performing model based on validation metrics. Configuration management used Hydra [47] to enable reproducible experimentation.

878

879 *Model evaluation*

Performance was monitored using precision, recall, and F1 score on the validation set after each epoch:

882

883

884

885

886

887

$$\begin{aligned}\text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, & \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{F1} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\end{aligned}$$

where TP (true positives) are chimeric reads correctly classified as artificial, TN (true negatives) are biological reads correctly classified as biological, FP (false positives) are biological reads misclassified as artificial, and FN (false negatives) are artificial reads misclassified as biological. Final model selection was based on best validation performance as determined by early stopping.

893

894 **Model inference and application**

895

896

Inference pipeline

To apply ChimeraLM to new WGA sequencing data, the model takes a BAM file as input. Chimeric reads are identified using SA tags and filtered to exclude unmapped, secondary, or supplementary alignments. Each chimeric read sequence is tokenized using the tokenizer (maximum length 32,768 bp, with truncation or padding as needed). The trained model processes sequences in batches, generating two logits $[z_0, z_1]$ for each read corresponding to biological and artificial chimeric classes. Classification is determined by $\hat{y} = \text{argmax}(z_0, z_1)$. ChimeraLM outputs a filtered BAM file containing only reads classified as biological, which can be directly used for downstream analyses including SV calling.

906

907 **Performance evaluation**

908

909

Test set evaluation

Final model performance was evaluated on the held-out test set and the independent MinION Mk1c dataset. Metrics (precision, recall, and F1 score) were computed as described in the training section, where true positives represent chimeric reads correctly classified as artificial and true negatives represent biological reads correctly classified as biological.

915

916

Read-length analysis

For every chimeric read (primary alignment with an SA tag) in the WGA PromethION and MinION BAM files, we recorded the full read length (including hard-clipped bases), the number of alignment segments, and the read-coordinate position of every chimeric junction, defined as the boundary between consecutive alignment segments

ordered along the read (primary alignment from its CIGAR string; supplementary alignments from the SA tag). A read was considered fully truncated if all of its junctions lay beyond position 32,768. Reads were grouped into length bins (≤ 2 kb, 2–8 kb, 8–16 kb, 16–32 kb, > 32 kb) and, for each bin, we computed the fraction of reads classified as artifacts and, for WGA-derived reads of the held-out test set, precision, recall and F1 using the ChimeraLM predictions generated for the full WGA datasets.

SV calling

SVs were called using multiple tools to ensure comprehensive detection. For long-read data (ONT PromethION P2 and MinION Mk1c), we used Sniffles v2.5 [14, 15], DeBreak v1.2 [16], SVIM v2.0.0 [17], and cuteSV v2.1.1 [18]. For short-read data of the PC3 cell line, we used both the CCLE Illumina WGS dataset and the PRJNA361315 Illumina WGS dataset, processed with Manta v1.6.0 [48], DELLY v1.5.0 [49], and SvABA v1.1.0 [50]. All tools were executed with default recommended parameters.

Gold standard SV dataset construction

To evaluate the impact of ChimeraLM on SV detection accuracy, we generated a high-confidence gold-standard SV set from bulk PC3 sequencing data. All SV comparisons and breakpoint corrections were performed using OctopusSV v0.2.3 [51]. Four bulk datasets were integrated: ONT MinION Mk1c, ONT PromethION P2, the CCLE Illumina WGS dataset, and the PRJNA361315 Illumina WGS dataset. SVs were called independently within each dataset, and events supported by at least two SV callers were retained. The remaining calls were then compared across datasets, and SVs observed in at least two of the four datasets were designated as gold-standard events for benchmarking. The sensitivity of the gold standard to this threshold was assessed by merging the four caller-consensus call sets with OctopusSV (`--union --mode sample`) and counting events supported by at least one, two, three or four datasets (Extended Data Fig. 4).

SV benchmarking analysis

To assess the impact of ChimeraLM on SV calling accuracy, we compared SV calls from unfiltered WGA data and ChimeraLM-filtered WGA data against two references: (1) the stringent multi-platform gold standard dataset, and (2) platform-matched long-read bulk sequencing data. Benchmarking was performed using Truvari v4.2.2 [52] with default parameters. SVs were considered supported if they matched reference variants within the defined breakpoint tolerance. Validation rates were calculated as the proportion of called SVs supported by the reference. This dual benchmarking strategy quantifies both improvements in detecting high-confidence multi-platform SVs and the retention of platform-specific true variants. For Fig. 3b–f, calls with fewer than three supporting reads (Sniffles2 `SUPPORT < 3`) were removed from the bulk, WGA and WGA + ChimeraLM call sets, matching the Sniffles2 default minimum read support (`--minsupport 3`); counts of supported and unsupported calls in Fig. 3b–e are the Truvari TP and FP counts, respectively.

External validation on HCC78

Raw ONT reads of the HCC78 single-cell WGA and bulk libraries of Ni et al. [34] were obtained from NCBI SRA (BioProject PRJNA875576; runs SRR21397273, MDA single cell, R10.4; SRR21397268, MALBAC single cell, R9.4.1; SRR21397275 and SRR21397274, bulk, R10.4 and R9.4.1) and processed exactly as the PC3 data. Chimeric reads were identified by SA tags as above, bulk-support labels were assigned by matching every WGA chimeric read against the chimeric reads of both bulk libraries with the same procedure and 1-kb tolerance used for PC3, and the released ChimeraLM model was applied with `chimeraalm predict` (batch size 12) without modification. Precision, recall and F1 were computed with artifacts as the positive class.

Benchmarking against existing methods

ChimeraLM was compared to existing computational methods for detecting amplification-induced chimeric artifacts: SACRA [25] (GitHub commit 9a2607e), 3rd-ChimeraMiner [13] (GitHub commit 04b5233), and CADECT v1.2 [11]. Both tools were applied to WGA data from PromethION P2 and MinION Mk1c platforms using default parameters as recommended in their documentation. Performance was evaluated by measuring the percentage reduction in chimeric reads relative to unprocessed WGA data. Chimeric reads were identified using WGA tag-based alignment criteria (reads with SA tags indicating split alignments), and reduction rates were calculated as the proportion of chimeric reads removed by each method.

Attention weight analysis

To investigate ChimeraLM’s interpretability, we analyzed attention weights from the pooling mechanism for representative chimeric reads. Attention weights indicate the relative importance assigned to each sequence position during classification. For selected reads, we extracted per-position attention weights and visualized them alongside read alignments to identify whether the model focuses on mechanistically relevant regions.

Chimeric junction positions were identified from alignment data (defined by breakpoints in SA tags). A region of ± 50 bp surrounding each junction was designated as the junction region. Attention weights within junction region were compared to non-junction regions using the Wilcoxon rank-sum test [53], with statistical significance assessed at $p < 0.001$.

Data visualization

Figures were generated using Python with Matplotlib [54] and Seaborn [55].

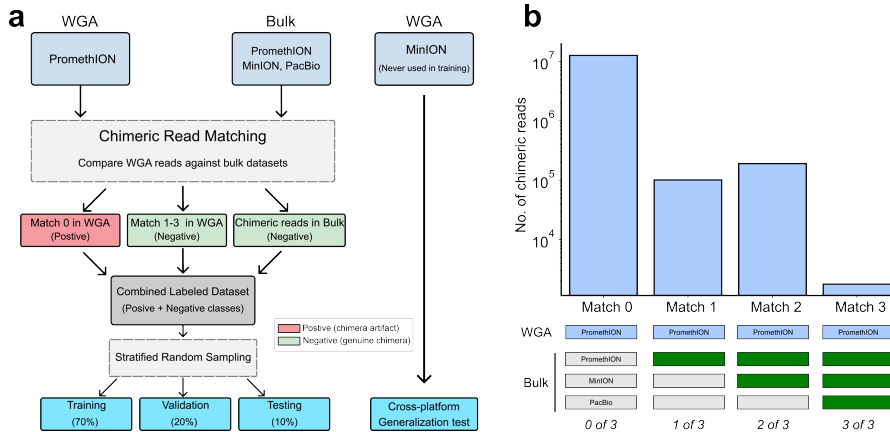
Computing resources

Computations were performed on a high performance computing (HPC) server with 64-core Intel Xeon Gold 6338 CPU, 256 GB RAM, and two NVIDIA A100 GPUs (80 GB memory each). Inference cost scales approximately linearly with input length: on

one A100 (bf16, batch size 12) the released model classifies 1,512 reads s⁻¹ at 2,048 bp, 402 reads s⁻¹ at 8,192 bp and 96 reads s⁻¹ at the 32,768-bp limit, with peak GPU memory of 0.4, 1.4 and 5.6 GB, respectively. For comparison, a HyenaDNA-medium backbone evaluated at its full 160,000-bp context processes 8 reads s⁻¹ and requires 31 GB at the same batch size.

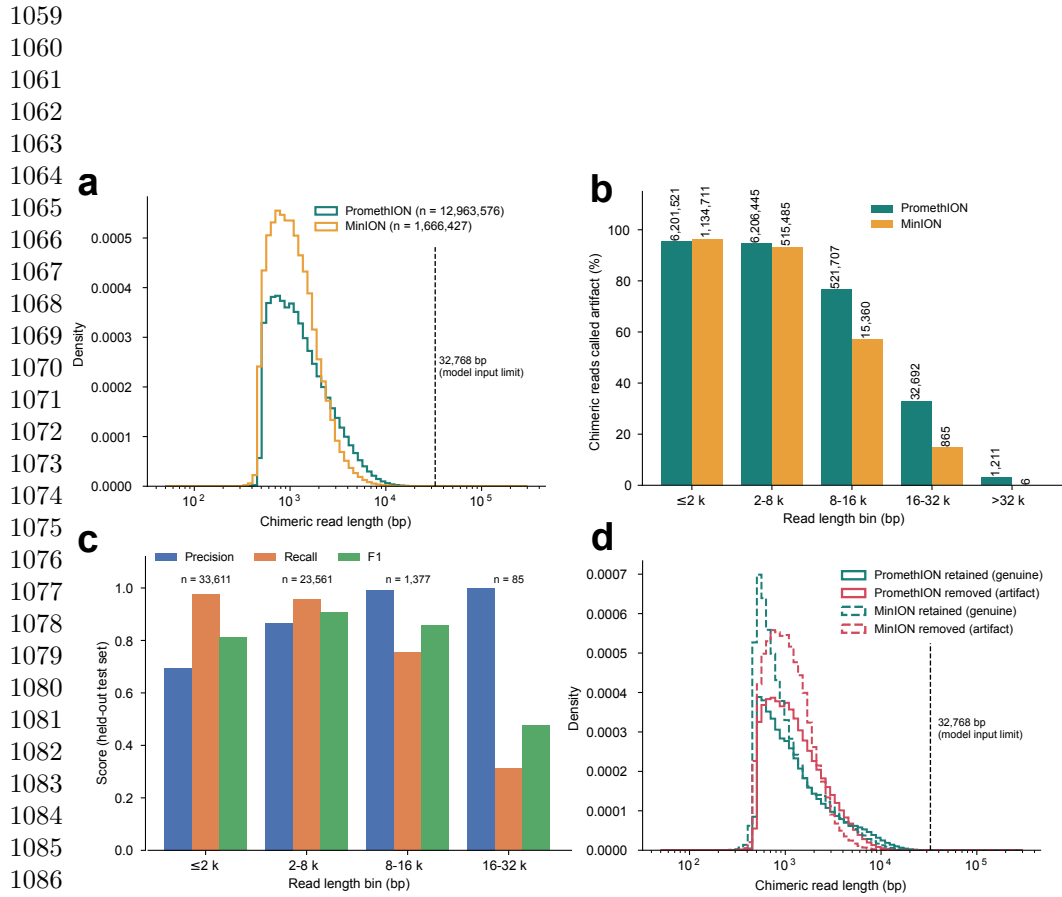
Extended Data Table 1 Sequencing and alignment statistics of PC3

Sample	Platform	Reads ($\times 10^6$)	Total bases (Gb)	Total bases aligned (Gb)	Fraction aligned	Mean length (bp)	Mean quality (Q)	Average identity (%)
WGA	MinION	9.11	14.6	10.4	0.7	1,603	14.3	97.6
WGA	PromethION	44.69	128.2	69.2	0.5	2,869	14.5	96.1
Bulk	MinION	0.97	8.1	7.1	0.9	8,310	17.2	97.3
Bulk	PromethION	8.00	69.9	62.4	0.9	8,732	18.5	97.7

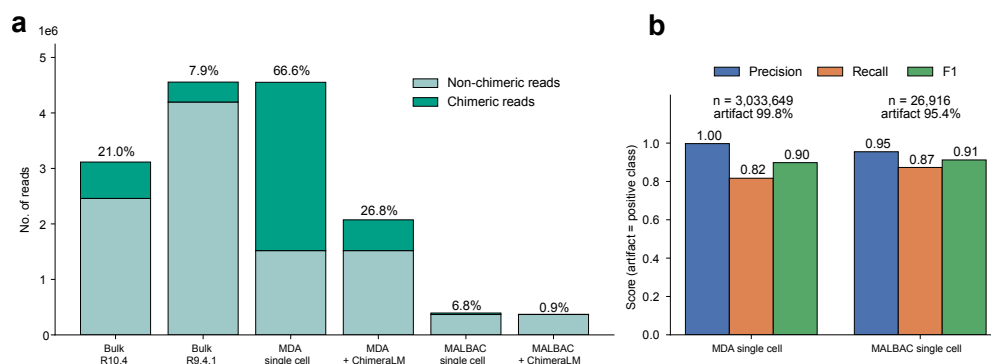


Extended Data Fig. 1 Training dataset construction and bulk-supported labeling strategy. (a) Workflow for generating labeled training data. WGA PromethION data is compared against three independent bulk sequencing datasets (PromethION, MinION, and PacBio). Reads with no bulk matches (Match 0) are labeled artificial; reads matching one or more bulk datasets (Match 1–3) are labeled biological, along with chimeric reads sampled directly from bulk data. The labeled dataset is split into training (70%), validation (20%), and test (10%) sets. The WGA MinION dataset is reserved for independent cross-platform evaluation. (b) Distribution of chimeric read matches. Bar chart shows the number of WGA PromethION chimeric reads (log scale) by bulk dataset matches. Match 0 reads (~10⁷) lacking bulk validation are classified as artificial; Match 1–3 reads with bulk support are classified as biological. The substantial imbalance reflects high prevalence of WGA-induced artifacts.

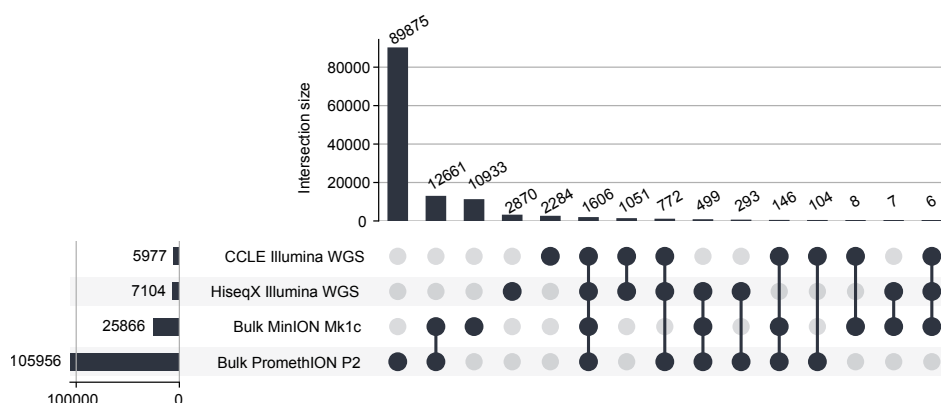
Supplementary information.



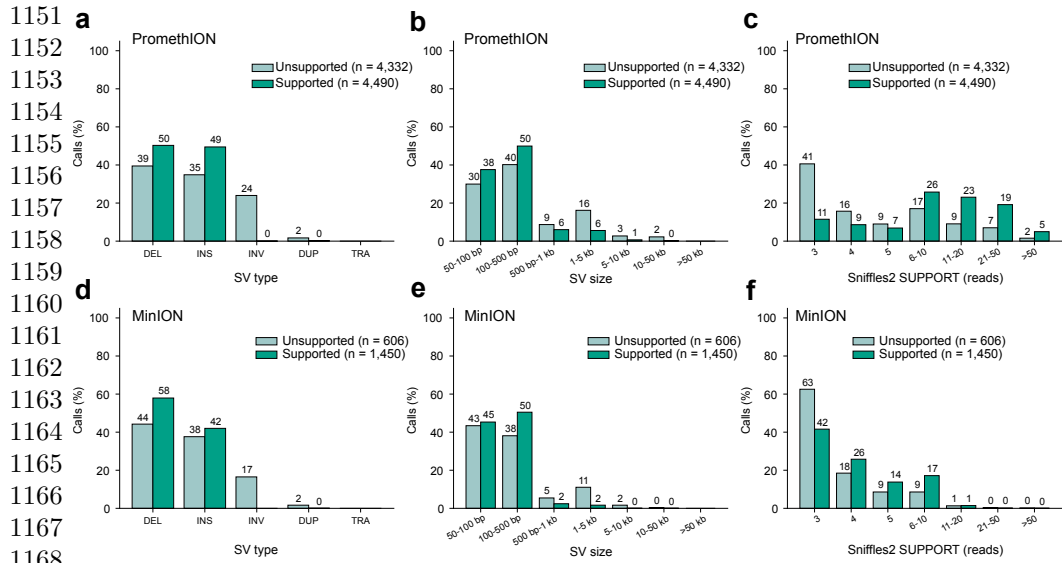
Extended Data Fig. 2 Read-length distribution of chimeric reads and length dependence of ChimeraLM classification. (a) Length distribution of chimeric reads (primary alignments carrying an SA tag) in the WGA PromethION ($n = 12,963,576$; median 2,100 bp, N50 4,053 bp, maximum 126,974 bp) and WGA MinION ($n = 1,666,427$; median 1,469 bp, N50 2,333 bp, maximum 42,118 bp) datasets. The dashed line marks the 32,768-bp model input limit, exceeded by 1,211 (0.009%) and 6 reads, respectively; for all but two of these reads at least one chimeric junction lies within the first 32,768 bp. (b) Fraction of chimeric reads classified as WGA-induced artifacts by read-length bin; numbers above bars give the number of reads per bin. (c) Precision, recall and F1 of ChimeraLM on WGA-derived held-out test reads ($n = 58,636$; the 17,876 test reads sampled from bulk data are not shown because they are not part of the WGA dataset) stratified by read length. Bins with fewer than 10 reads are omitted. (d) Length distributions of chimeric reads retained (classified as genuine) and removed (classified as artifact) by ChimeraLM on both platforms (PromethION retained: median 3,087 bp, N50 7,989 bp; removed: median 2,067 bp, N50 3,836 bp; MinION retained: median 2,031 bp, N50 5,560 bp; removed: median 1,458 bp, N50 2,220 bp).



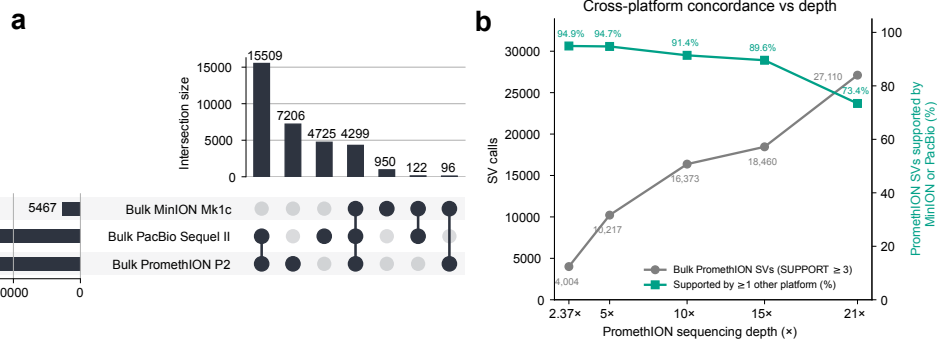
Extended Data Fig. 3 External validation of the trained model on single-cell WGA data of an independent cell line (HCC78). Data from Ni et al. [34] (NCBI BioProject PRJNA875576): single HCC78 cells amplified with REPLI-g Single Cell MDA (R10.4 flow cell) or MALBAC (R9.4.1), and two unamplified bulk libraries (R10.4, R9.4.1). The PC3-trained ChimeraLM model was applied without retraining. **(a)** Chimeric (dark teal) and non-chimeric (light teal) mapped reads with the chimeric fraction above each bar, for the bulk libraries, the WGA libraries, and the WGA libraries after ChimeraLM filtering (MDA: 66.6% to 26.8%; MALBAC: 6.8% to 0.9%). **(b)** Precision, recall and F1 for the artifact class against bulk-support labels derived from the two HCC78 bulk libraries (same matching procedure as for PC3); *n* is the number of labelled chimeric reads and the percentage labelled artifact is given.



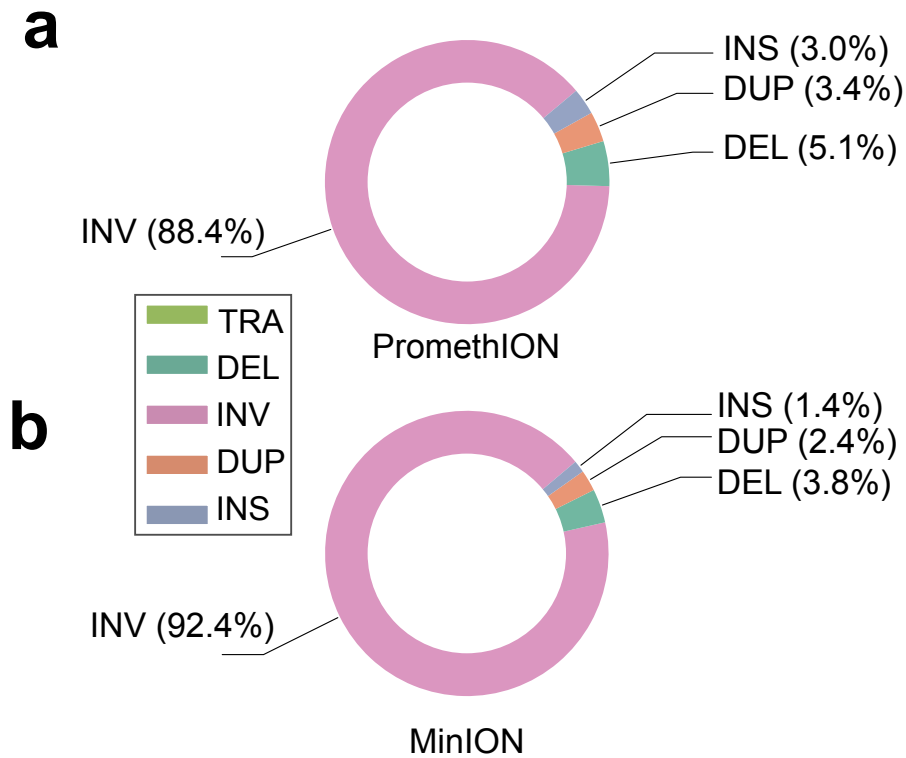
Extended Data Fig. 4 Overlap of structural variant calls among the four bulk PC3 datasets used to build the gold standard. UpSet plot of caller-consensus SVs (≥ 2 callers per dataset) from bulk ONT MinION Mk1c (25,866 calls), bulk ONT PromethION P2 (105,956), CCLE Illumina WGS (5,977) and PRJNA361315 (HiSeq X) Illumina WGS (7,104), merged with OctopusV. The union contains 123,115 events; 17,153 are supported by ≥ 2 datasets (the gold standard used in Fig. 3), 3,029 by ≥ 3 and 1,606 by all four. Within the ≥ 2 set, 12,661 events (73.8%) are shared only by the two long-read datasets, 3,441 (20.1%) have both long- and short-read support and 1,051 (6.1%) are shared only by the two short-read datasets.



Extended Data Fig. 5 Features of structural variant calls that remain unsupported after ChimeraLM filtering. SV calls from the WGA + ChimeraLM call sets (Sniffles2, ≥ 3 supporting reads) benchmarked against the gold standard (Fig. 3b,c) were split into unsupported (light teal; Truvari false positives) and supported (dark teal; Truvari true positives) calls. (a–c) PromethION ($n = 4,332$ unsupported, 4,490 supported): distribution of SV type (a), SV size (b) and Sniffles2 read support (c). (d–f) The same for MinION ($n = 606$ unsupported, 1,450 supported). Numbers above bars give the percentage of calls in each class.



Extended Data Fig. 6 Concordance of structural variant calls among the three bulk PC3 long-read datasets. (a) UpSet plot of high-confidence SVs (Sniffles2, ≥ 3 supporting reads; merged with OctopusSV) from bulk PromethION P2 (20.8x; 27,110 calls), bulk MinION Mk1c (2.4x; 5,467) and bulk PacBio Sequel II (38x; 24,655). Of the 32,907 union calls, 20,026 (60.9%) are detected on at least two platforms and 4,299 (13.1%) on all three; platform-private calls make up 26.6%, 17.4% and 19.2% of the PromethION, MinION and PacBio call sets, respectively. (b) Depth titration of the bulk PromethION data (down-sampled to 2.37x, 5x, 10x, 15x and full 21x): number of PromethION SV calls (grey) and the fraction supported by at least one of the other two platforms (teal). The platform-private fraction rises with depth (5.1% to 26.6%), showing that deeper sequencing recovers variants absent from the shallower reference sets.



Extended Data Fig. 7 Composition of artifact-supported SVs for PromethION (a) and MinION (b). Donut charts summarize the distribution of SV types including translocation (TRA), deletion (DEL), inversion (INV), duplication (DUP), and insertion (INS) among events supported exclusively by chimera artifacts, representing artificial SVs preferentially removed by ChimeraLM(all Sniffles2 calls, without a minimum read-support filter).

Acknowledgements. We thank Tingyou Wang for guidance on figure preparation. This project was supported in part by NIH grants R35GM142441 and R01CA259388 awarded to RY.

Declarations

Author Contributions. YL, QG and RY designed the study. YL and QG performed the analysis. QG performed the experiments. YL and QG designed and implemented the model. YL built the command-line tool and documentation. YL, QG and RY wrote the manuscript. RY supervised this work.

Data Availability. The raw sequencing data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession PRJNA1354861. The dataset includes Oxford Nanopore long-read whole-genome sequencing of PC3 prostate cancer cells and MDA-amplified single-cell derivatives. The individual SRA accessions are as follows: PC3 bulk (MinION Mk1C), SRR35904028; PC3 bulk (PromethION P2), SRR35904029; PC3 10-cell WGA (MinION Mk1C), SRR35904026; PC3 10-cell WGA (PromethION P2), SRR35904027. We can access the data at the following link: <https://dataview.ncbi.nlm.nih.gov/object/PRJNA1354861?reviewer=viej6cv6mgbli3n7a9a5k1bsb3>

Code Availability. ChimeraLM, implemented in Python, is open source and available on GitHub (<https://github.com/ylab-hi/ChimeraLM>) under the Apache License, Version 2.0. The package can be installed via PyPI (<https://pypi.org/project/chimeralm>) using pip, with wheel distributions provided for Windows, Linux, and macOS to ensure easy cross-platform installation. An interactive demo is available on Hugging Face (<https://huggingface.co/spaces/yangliz5/ChimeraLM>), allowing users to test ChimeraLM’s functionality without local installation; the demo runs on free-tier hardware and is put to sleep after 48 hours without visitors, so it may take one to two minutes to restart when first opened. For large-scale analyses, we recommend using ChimeraLM on systems with GPU acceleration. Detailed system requirements and optimization guidelines are available in the repository’s documentation (<https://ylab-hi.github.io/ChimeraLM/>).

Conflict of interest. The authors declare no competing interests.

Acronyms

CPU central processing unit 14

DEL deletion 8, 27

dMDA droplet-based MDA 14

DUP duplication 8, 27

GLM genomic language model 1, 3, 15

GPU graphics processing unit 14, 19, 22, 23, 28

HPC high performance computing 22

INS insertion	8, 27	1289
INV inversion	8, 27	1290
		1291
LIANTI linear amplification via transposon insertion	14	1292
		1293
MALBAC multiple annealing and looping-based amplification cycles	7, 14, 22, 25	1294
MDA multiple displacement amplification	2, 7, 10, 22, 25	1295
MLP multilayer perceptron	4, 5, 18, 19	1296
		1297
ONT Oxford Nanopore Technologies	2, 3, 8–10, 13, 16, 21, 25	1298
		1299
PacBio Pacific Biosciences	2, 3, 16	1300
PTA primary template-directed amplification	14	1301
		1302
SA supplementary alignment	16, 20–22, 24	1303
SV structural variation	1–5, 8–11, 13–15, 17, 20, 21, 25–27	1304
		1305
TRA translocation	8, 27	1306
		1307
WGA whole genome amplification	1–10, 12–17, 20–26	1308
WGS whole genome sequencing	8, 9, 21, 25	1309

References

[1] Kalef-Ezra, E. <i>et al.</i> Single-cell somatic copy number variants in brain using different amplification methods and reference genomes. <i>Communications Biology</i> 1288 (2024).	1310
	1311
[2] Navin, N. <i>et al.</i> Tumour evolution inferred by single-cell sequencing. <i>Nature</i> 472 , 90–94 (2011).	1312
	1313
[3] Sun, C. <i>et al.</i> Mapping recurrent mosaic copy number variation in human neurons. <i>Nature Communications</i> 4220 (2024).	1314
	1315
[4] Gawad, C., Koh, W. & Quake, S. R. Single-cell genome sequencing: current state of the science. <i>Nature Reviews Genetics</i> 175–188 (2016).	1316
	1317
[5] Chen, C. <i>et al.</i> Single-cell whole-genome analyses by linear amplification via transposon insertion (LIANTI). <i>Science (new York, N.Y.)</i> 356 , 189–194 (2017).	1318
	1319
[6] Macaulay, I. C. & Voet, T. Single cell genomics: Advances and future perspectives. <i>PLOS Genetics</i> 10 , e1004126 (2014).	1320
	1321
[7] de Bourcy, C. F. A. <i>et al.</i> A quantitative comparison of single-cell whole genome amplification methods. <i>PLoS ONE</i> e105585 (2014).	1322
	1323
[8] Biezuner, T. <i>et al.</i> Comparison of seven single cell whole genome amplification commercial kits using targeted sequencing. <i>Scientific Reports</i> 17171 (2021).	1324
	1325
	1326
	1327
	1328
	1329
	1330
	1331
	1332
	1333
	1334

1335 [9] Lu, N., Qiao, Y., Lu, Z. & Tu, J. Chimera: The spoiler in multiple displacement
1336 amplification. *Computational and Structural Biotechnology Journal* 1688–1696
1337 (2023).
1338

1339 [10] Lasken, R. S. & Stockwell, T. B. Mechanism of chimera formation during the
1340 multiple displacement amplification reaction. *BMC Biotechnology* **7**, 19 (2007).
1341

1342 [11] Agyabeng-Dadzie, F. *et al.* Evaluating the benefits and limits of multiple displace-
1343 ment amplification with whole-genome oxford nanopore sequencing. *Molecular*
1344 *Ecology Resources* e14094 (2025).

1345 [12] Dean, F. B. *et al.* Comprehensive human genome amplification using multiple
1346 displacement amplification. *Proceedings of the National Academy of Sciences* **99**,
1347 5261–5266 (2002).
1348

1349 [13] Lu, N. *et al.* Exploration of whole genome amplification generated chimeric
1350 sequences in long-read sequencing data. *Briefings in Bioinformatics* **24**, bbad275
1351 (2023).
1352

1353 [14] Sedlazeck, F. J. *et al.* Accurate detection of complex structural variations using
1354 single-molecule sequencing. *Nature Methods* 461–468 (2018).
1355

1356 [15] Smolka, M. *et al.* Detection of mosaic and population-level structural variants
1357 with sniffles2. *Nature Biotechnology* 1571–1580 (2024).
1358

1359 [16] Chen, Y. *et al.* Deciphering the exact breakpoints of structural variations using
1360 long sequencing reads with DeBreak. *Nature Communications* 283 (2023).
1361

1362 [17] Heller, D. & Vingron, M. SVIM: Structural variant identification using mapped
1363 long reads. *Bioinformatics* 2907–2915 (2019).
1364

1365 [18] Jiang, T. *et al.* Long-read-based human genomic structural variation detection
1366 with cuteSV. *Genome Biology* 189 (2020).

1367 [19] Gonzalez-Pena, V. *et al.* Accurate genomic variant detection in single cells with
1368 primary template-directed amplification. *Proceedings of the National Academy of*
1369 *Sciences* **118**, e2024176118 (2021).
1370

1371 [20] Kosugi, S. *et al.* Comprehensive evaluation of structural variation detection
1372 algorithms for whole genome sequencing. *Genome Biology* **20**, 117 (2019).
1373

1374 [21] Alkan, C., Coe, B. P. & Eichler, E. E. Genome structural variation discovery and
1375 genotyping. *Nature Reviews Genetics* **12**, 363–376 (2011).
1376

1377 [22] Hård, J. *et al.* Long-read whole-genome analysis of human single cells. *Nature*
1378 *Communications* **14**, 5164 (2023).
1379
1380

[23]	Gupta, P., O'Neill, H., Wolvetang, E. J., Chatterjee, A. & Gupta, I. Advances in single-cell long-read sequencing technologies. <i>NAR Genomics and Bioinformatics</i> 6 , lqae047 (2024).	1381 1382 1383 1384
[24]	Wen, L. & Tang, F. Single-cell omics sequencing technologies: The long-read generation. <i>Trends in Genetics</i> 0 (2025).	1385 1386 1387
[25]	Kiguchi, Y., Nishijima, S., Kumar, N., Hattori, M. & Suda, W. Long-read metagenomics of multiple displacement amplified DNA of low-biomass human gut phageomes by SACRA pre-processing chimeric reads. <i>DNA Research</i> 28 , dsab019 (2021).	1388 1389 1390 1391
[26]	Ha, Y.-J. <i>et al.</i> Comprehensive benchmarking and guidelines of mosaic variant calling strategies. <i>Nature Methods</i> 20 , 2058–2067 (2023).	1392 1393 1394
[27]	Höijer, I. <i>et al.</i> Amplification-free long-read sequencing reveals unforeseen CRISPR-Cas9 off-target activity. <i>Genome Biology</i> 21 , 290 (2020).	1395 1396 1397
[28]	Li, Y. <i>et al.</i> A genomic language model for chimera artifact detection in nanopore direct rna sequencing. <i>bioRxiv</i> (2024). URL https://www.biorxiv.org/content/early/2024/10/25/2024.10.23.619929 .	1398 1399 1400 1401
[29]	Nguyen, E. <i>et al.</i> <i>HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution</i> , Vol. 36, 43177–43201 (Curran Associates, Inc., 2023).	1402 1403 1404
[30]	Dalla-Torre, H. <i>et al.</i> Nucleotide transformer: building and evaluating robust foundation models for human genomics. <i>Nature Methods</i> 287–297 (2025).	1405 1406 1407
[31]	Zhou, Z. <i>et al.</i> <i>DNABERT-2: Efficient foundation model and benchmark for multi-species genomes</i> , 1–24 (OpenReview.net, 2024).	1408 1409
[32]	Consens, M. E. <i>et al.</i> To transformers and beyond: Large language models for the genome (2023). arXiv:2311.07621 .	1410 1411 1412
[33]	Poli, M. <i>et al.</i> <i>Hyena hierarchy: Towards larger convolutional language models</i> , Vol. 202, 28043–28078 (PMLR, 2023).	1413 1414 1415
[34]	Ni, Y., Liu, X., Simeneh, Z. M., Yang, M. & Li, R. Benchmarking of Nanopore R10.4 and R9.4.1 flow cells in single-cell whole-genome amplification and whole-genome shotgun sequencing. <i>Computational and Structural Biotechnology Journal</i> 21 , 2352–2364 (2023).	1416 1417 1418 1419 1420
[35]	Barretina, J. <i>et al.</i> The Cancer Cell Line Encyclopedia enables predictive modelling of anticancer drug sensitivity. <i>Nature</i> 483 , 603–607 (2012).	1421 1422 1423
[36]	Seim, I., Jeffery, P. L., Thomas, P. B., Nelson, C. C. & Chopin, L. K. Whole-Genome Sequence of the Metastatic PC3 and LNCaP Human Prostate Cancer Cell Lines. <i>G3 (Bethesda, Md.)</i> 7 , 1731–1741 (2017).	1424 1425 1426

1427 [37] Mahmoud, M. *et al.* Structural variant calling: The long and the short of it.
1428 *Genome Biology* **20**, 246 (2019).
1429

1430 [38] Dippenaar, A. *et al.* Droplet based whole genome amplification for sequencing
1431 minute amounts of purified mycobacterium tuberculosis DNA. *Scientific Reports*
1432 **14**, 9931 (2024).
1433

1434 [39] PLC., O. N. Dorado. <https://github.com/nanoporetech/dorado> (2023).
1435

1436 [40] Martin, M. Cutadapt removes adapter sequences from high-throughput sequenc-
1437 ing reads. *Embnnet.journal* **17**, 10–12 (2011).
1438

1439 [41] Li, H. Minimap2: Pairwise alignment for nucleotide sequences. *Bioinformatics*
1440 **30**, 3094–3100 (2018).
1441

1442 [42] Danecek, P. *et al.* Twelve years of SAMtools and BCFtools. *GigaScience* giab008
1443 (2021).
1444

1445 [43] De Coster, W. & Rademakers, R. NanoPack2: Population-scale evaluation of
1446 long-read sequencing data. *Bioinformatics* **39**, btad311 (2023).
1447

1448 [44] Paszke, A. *et al.* *PyTorch: An imperative style, high-performance deep learning*
1449 *library*, Vol. 32, 8024–8035 (Curran Associates, Inc., 2019).
1450

1451 [45] Falcon, W. & The PyTorch Lightning team. PyTorch Lightning. GitHub
1452 repository (2019). URL <https://github.com/Lightning-AI/lightning>.
1453

1454 [46] Loshchilov, I. & Hutter, F. *Decoupled weight decay regularization* (2019).
1455

1456 [47] Yadan, O. Hydra - a framework for elegantly configuring complex applications.
1457 GitHub repository (2019). URL <https://github.com/facebookresearch/hydra>.
1458

1459 [48] Chen, X. *et al.* Manta: Rapid detection of structural variants and indels for
1460 germline and cancer sequencing applications. *Bioinformatics* **32**, 1220–1222 (2016).
1461

1462 [49] Rausch, T. *et al.* DELLY: Structural variant discovery by integrated paired-end
1463 and split-read analysis. *Bioinformatics* **28**, i333–i339 (2012).
1464

1465 [50] Wala, J. A. *et al.* SvABA: Genome-wide detection of structural variants and
1466 indels by local assembly. *Genome Research* **28**, 581–591 (2018).
1467

1468 [51] Guo, Q., Li, Y., Wang, T.-Y., Ramakrishnan, A. & Yang, R. OctopusSV and
1469 TentacleSV: A one-stop toolkit for multi-sample, cross-platform structural variant
1470 comparison and analysis. *Bioinformatics* **39**, btaf599 (2025).
1471

1472 [52] English, A. C., Menon, V. K., Gibbs, R. A., Metcalf, G. A. & Sedlazeck, F. J.
Truvari: Refined structural variant comparison preserves allelic diversity. *Genome*
Biology **23**, 271 (2022).

[53]	Virtanen, P. <i>et al.</i> SciPy 1.0: Fundamental algorithms for scientific computing in python. <i>Nature Methods</i> 261–272 (2020).	1473 1474 1475
[54]	Hunter, J. D. Matplotlib: A 2d graphics environment. <i>Computing in Science & Engineering</i> 90–95 (2007).	1476 1477 1478
[55]	Waskom, M. L. seaborn: statistical data visualization. <i>Journal of Open Source Software</i> 3021 (2021).	1479 1480 1481 1482 1483 1484 1485 1486 1487 1488 1489 1490 1491 1492 1493 1494 1495 1496 1497 1498 1499 1500 1501 1502 1503 1504 1505 1506 1507 1508 1509 1510 1511 1512 1513 1514 1515 1516 1517 1518